Final Project Data 211

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Data Collection

Here I am going to define the base Data Frame I will be adding to over the course of these three weeks.

```
SongCue <- read_excel("C:/Users/avoio/OneDrive/Documents/Data211_HW/Data/Data 211 Project Data.xlsx", sh
FirstSongBase<-numeric(588)
SongCue["FirstSong"]<-FirstSongBase
#SongCue$TotalSubscribers<-as.numeric(SongCue$TotalSubscribers)
#SongCue$TotalViews<-as.numeric(SongCue$TotalViews)
#SongCue$IsLiked<-as.logical(SongCue$IsLiked)
#SongCue$IsSubscribed<-as.logical(SongCue$IsSubscribed)
#write.csv(SongCue, file = "FinalProjectData.csv")
```

Here I am going to create a function to parse a string of values and add based to the base data frame.

```
#uses a vector of 100 values
addNewSongCue <- function(songVector){
  #sifts through each value
  for(i in 1:length(songVector)){
    #Find the song in the base data frame
    #define index to failsafe the song not being in the vector.
    index<-0
    for(j in 1:588){
      if(tolower(SongCue$SongName[j]) == tolower(songVector[i])){
        #setting index to j when they find the right name
        #will end the loop and find the correct index
        #to add to.
        index=j
        break
      }
    }
    if(i==1 && index!=0){
      SongCue$FirstSong[index]<-SongCue$FirstSong[index]+1
    }
    if(i<=10 && index!=0){
      SongCue$InTopTen[index]<-SongCue$InTopTen[index]+1
    }
    if(i<=25 && index!=0){
      SongCue$InTopTwentyFive[index]<-
        SongCue$InTopTwentyFive[index]+1
    }
```

```
    if(i<=50 && index!=0) {
      SongCue$InTopFifty[index]<-SongCue$InTopFifty[index]+1
    }
    if(i<=100 && index!=0) {
      SongCue$InTopOneHundred[index]<-
        SongCue$InTopOneHundred[index]+1
    }
  }
  return(SongCue)
}
```

Here is going to be each vector I will add. I will report them in order of the list, and only the first 100 of each shuffle.

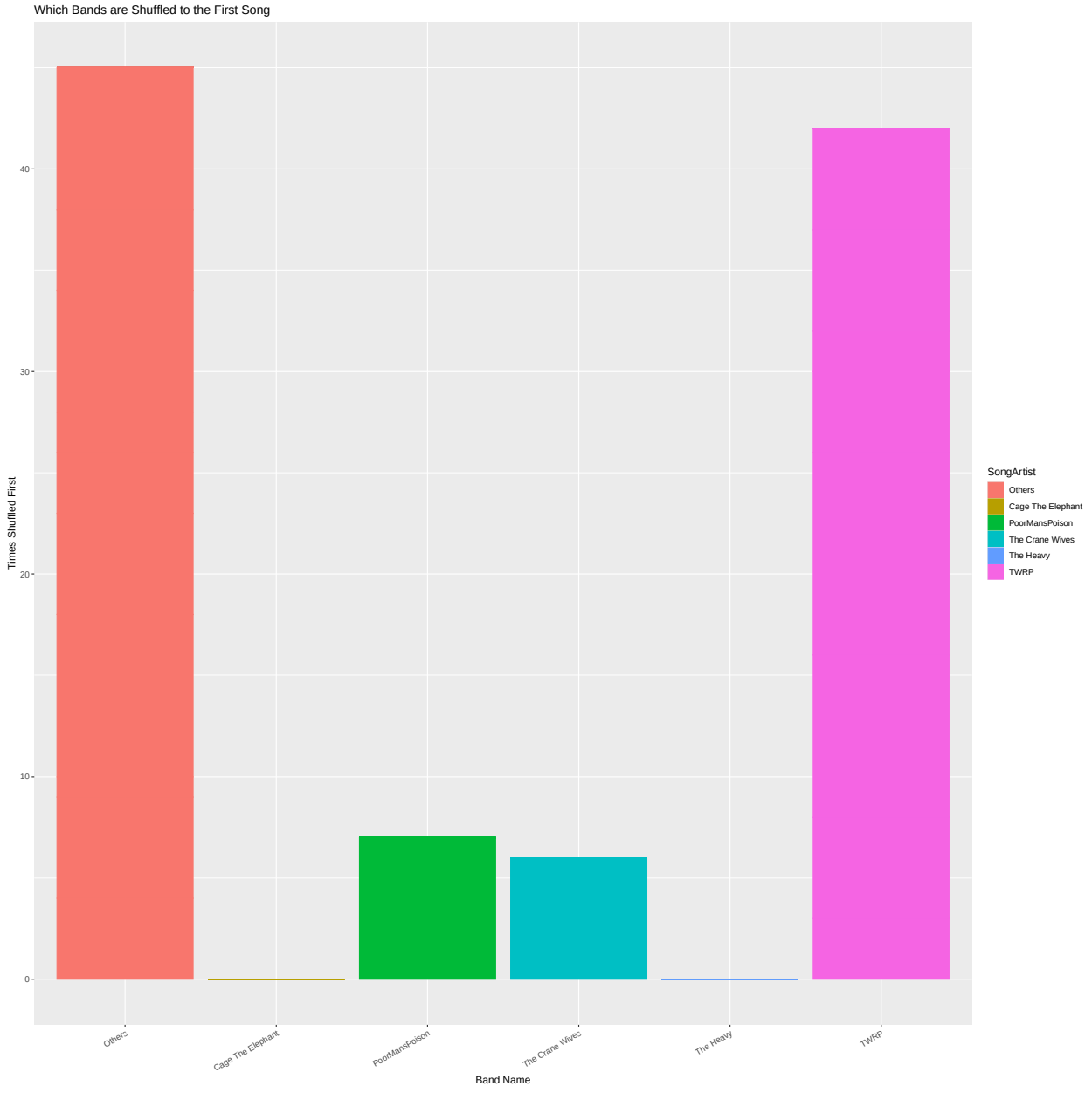
```
SongVectorList<-read_excel("C:/Users/avoio/OneDrive/Documents/Data211_HW/Data/Data 211 Project Data.xlsx
SongVectorList<-SongVectorList%>%
  select(-BaseTitles)%>%
  slice(-(101:588))
SongVectorList<-SongVectorList[,1:100]
for(i in 1:length(SongVectorList)){
  SongCue<-addNewSongCue(SongVectorList[[i]])
}
SongCue$SongArtist<-as.factor(SongCue$SongArtist)
SongCue$SongArtist<-fct_collapse(SongCue$SongArtist,
                                 Others = c("The Fray", "Good Kid", "RichaadEB",
                     "Adam Jones", "Barns Courtney", "Boy Pablo", "AJR", "Aviators",
                     "Bill Withers", "Billy Joel", "Burn The Ballroom", "Colter Wall",
                     "CyYu Music", "Des Rocs", "Disturbed", "Don McLean", "Dr. Villain",
                     "Dropkick Murphys", "Electric Swing Circus","ElipsisEditsYT",
                     "Fall Out Boy", "Finger Eleven", "Foo Fighters", "Friend With The Enemy",
                     "Fries On The Side", "Genesis", "Ghoultown","Godsmack", "GoldenBoys",
                     "Green Day", "half*alive", "IDKHOW", "Instrument Maniac", "Jonathan Young",
                     "Keith Power", "Kocchi no Kento", "Lemon Demon & Marina & The Diamonds",
                     "Lovejoy", "Maroon 5", "Matthew Raetzel", "MGMT", "Miike Snow",
                     "Minecraft", "MyNewSoundtrack", "NateWantsToBattle",
                     "Nejime Studios", "Nick Lutsko", "Night Runner", "Phil Collins",
                     "PUNPEE", "Rag'n'Bone Man", "Rob Cantor", "Sail North",
                     "Saint Motel", "Sam Tinnesz", "Savage Garden", "Seal", "Seo Jayeong",
                     "Shadow Academy", "Shayfer James", "Shinedown", "SIAMES",
                     "Simon & Garfunkel", "Smash Into Pieces", "Star-Lord", "Studio Yuraki",
                     "Super Guitar Bros", "Tally Hall", "Tears For Fears", "The Brother's Bright",
                     "The Brothers Bright", "The Dead South", "The Devil Makes Three", "The Fray",
                     "The Midnight", "The Offspring", "The Oh Hellos", "The Rolling Stones",
                     "The Score", "The Stupendium", "The Unlikely Candidates", "Three Days Grace",
                     "Weathers", "Weird Al Yankovic", "Will Stetson", "Younger Hunger", "Yumi Kawamura"))
SongCue$SongArtist<-fct_collapse(SongCue$SongArtist,
                                 PoorMansPoison = "PMP")
table(SongCue$SongArtist)
```

```
##
##            Others Cage The Elephant    PoorMansPoison   The Crane Wives
##               369                33                48                35
##         The Heavy              TWRP
```

```
##                 35                 68
```

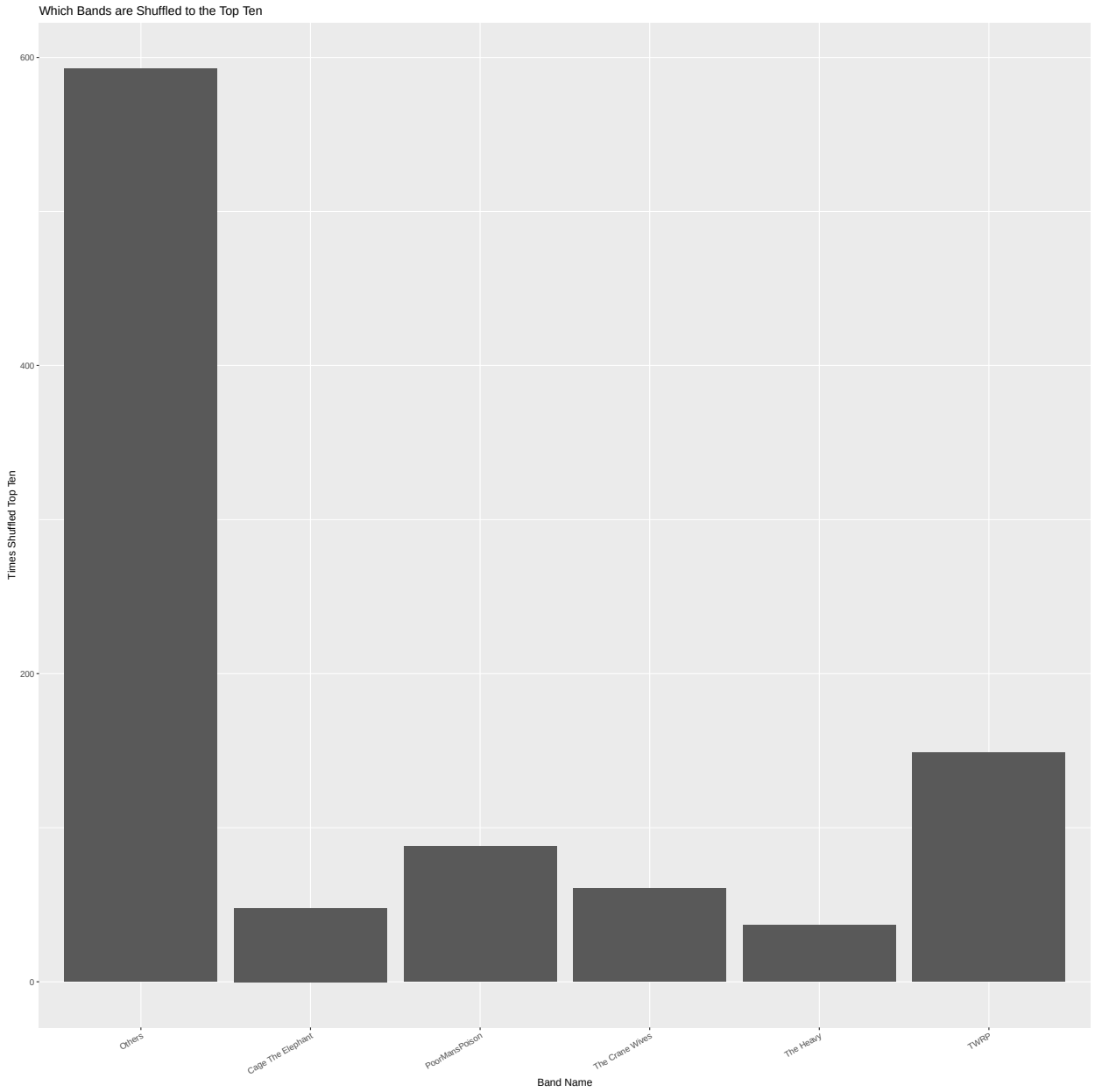
Here is where I will graph the placements vs the songs

```
ggplot(SongCue, aes(x=SongArtist,y=FirstSong))+
  geom_col(aes(fill=SongArtist,color=SongArtist))+
  theme(axis.text.x=element_text(hjust=1,angle=30))+
  labs(title = "Which Bands are Shuffled to the First Song",
       x = "Band Name", y = "Times Shuffled First")
```

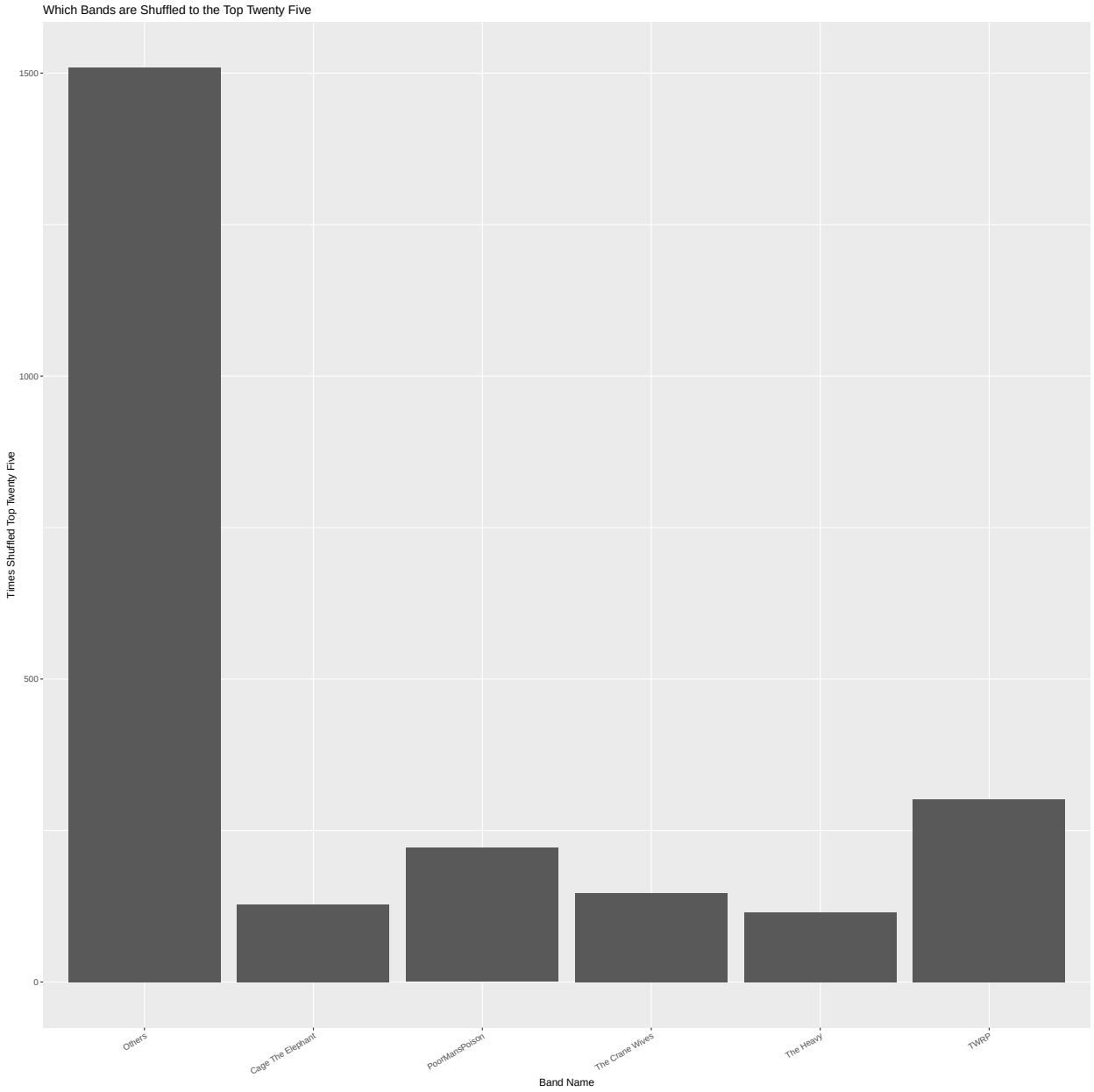



```
ggplot(SongCue, aes(x=SongArtist,y=InTopTen))+
  geom_col()+
  theme(axis.text.x=element_text(hjust=1,angle=30))+
```

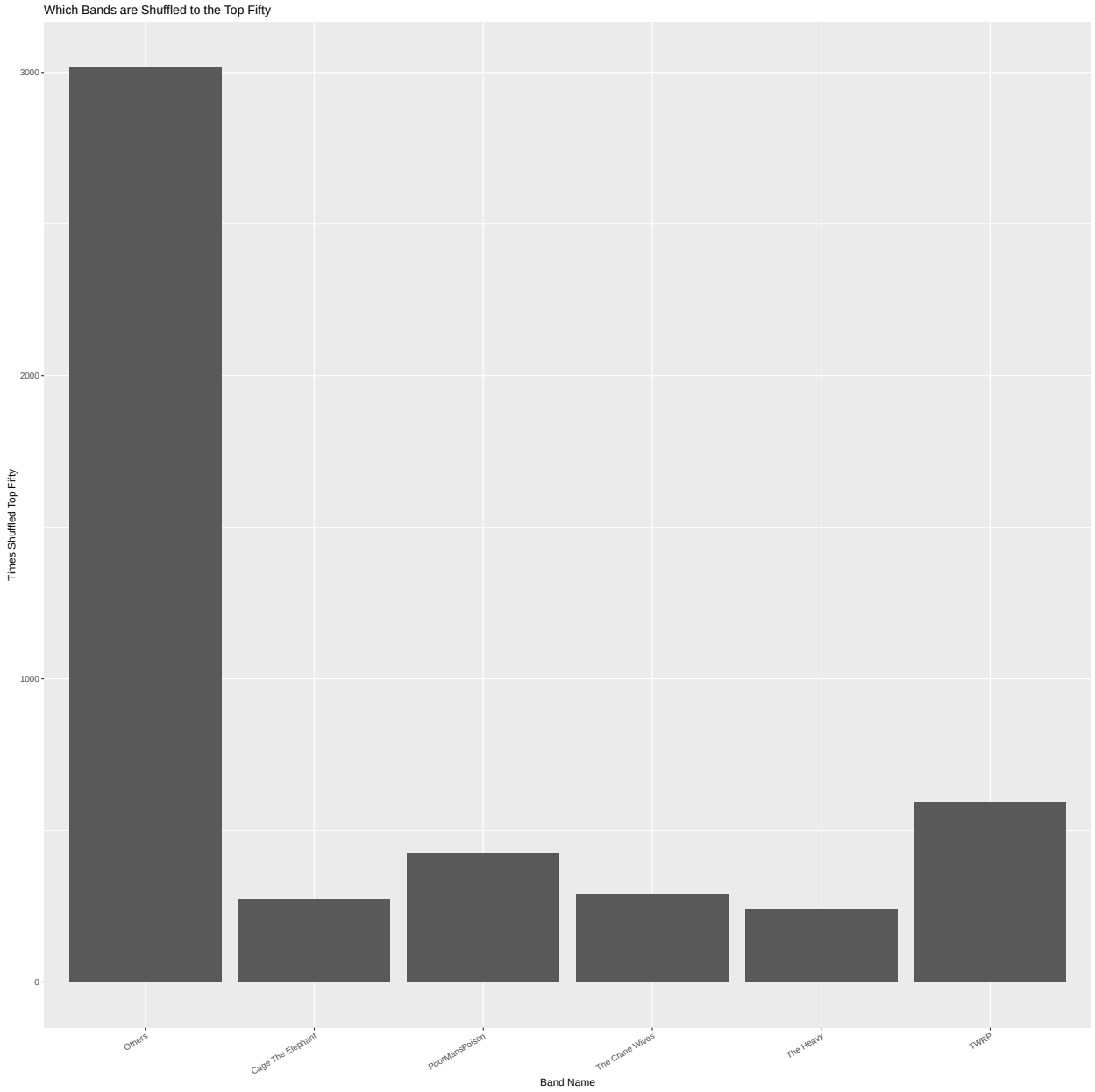
```
    labs(title = "Which Bands are Shuffled to the Top Ten",
       x = "Band Name", y = "Times Shuffled Top Ten")
```



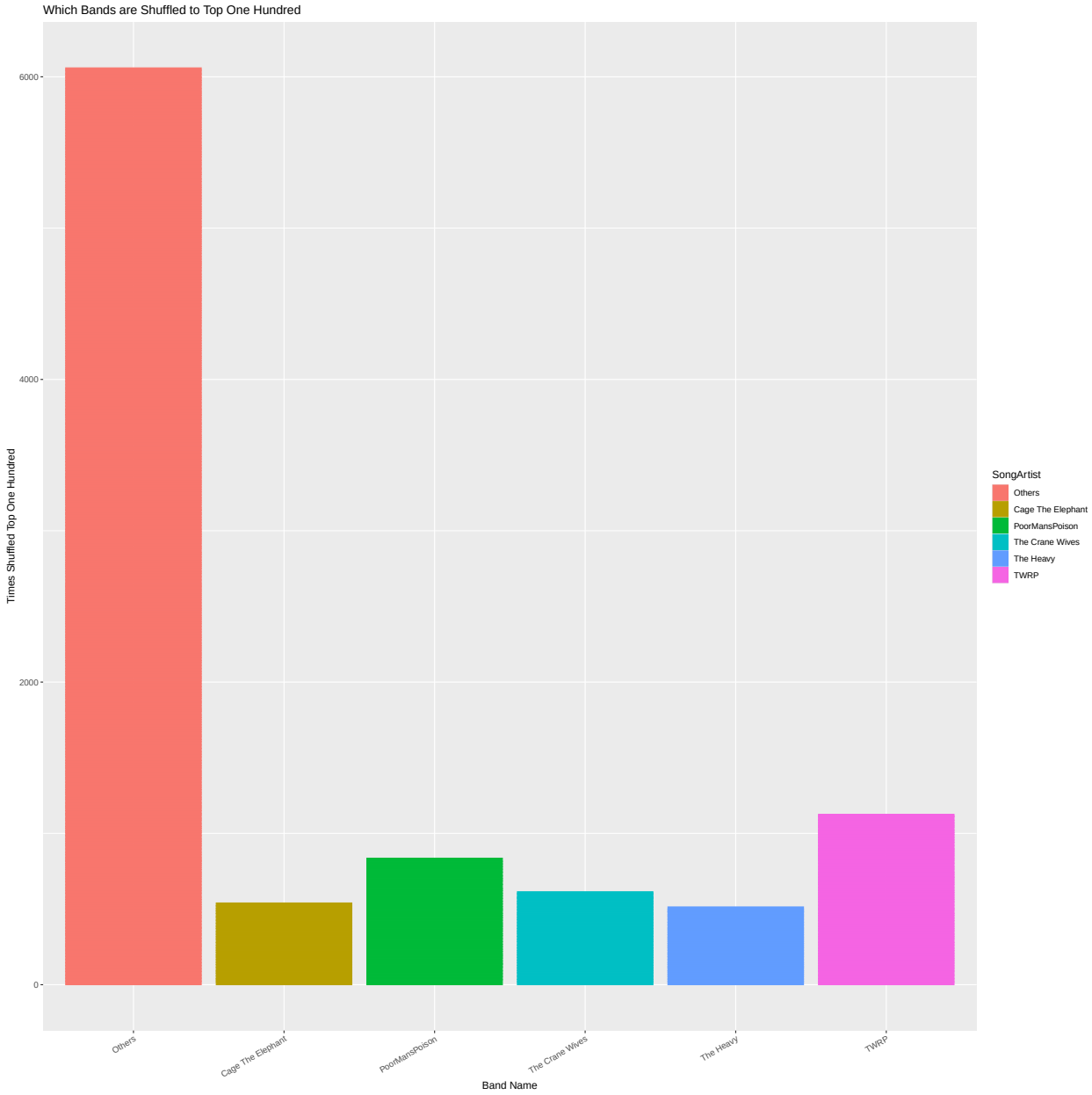

```
ggplot(SongCue, aes(x=SongArtist,y=InTopTwentyFive))+
  geom_col()+
  theme(axis.text.x=element_text(hjust=1,angle=30))+
    labs(title = "Which Bands are Shuffled to the Top Twenty Five",
       x = "Band Name", y = "Times Shuffled Top Twenty Five")
```



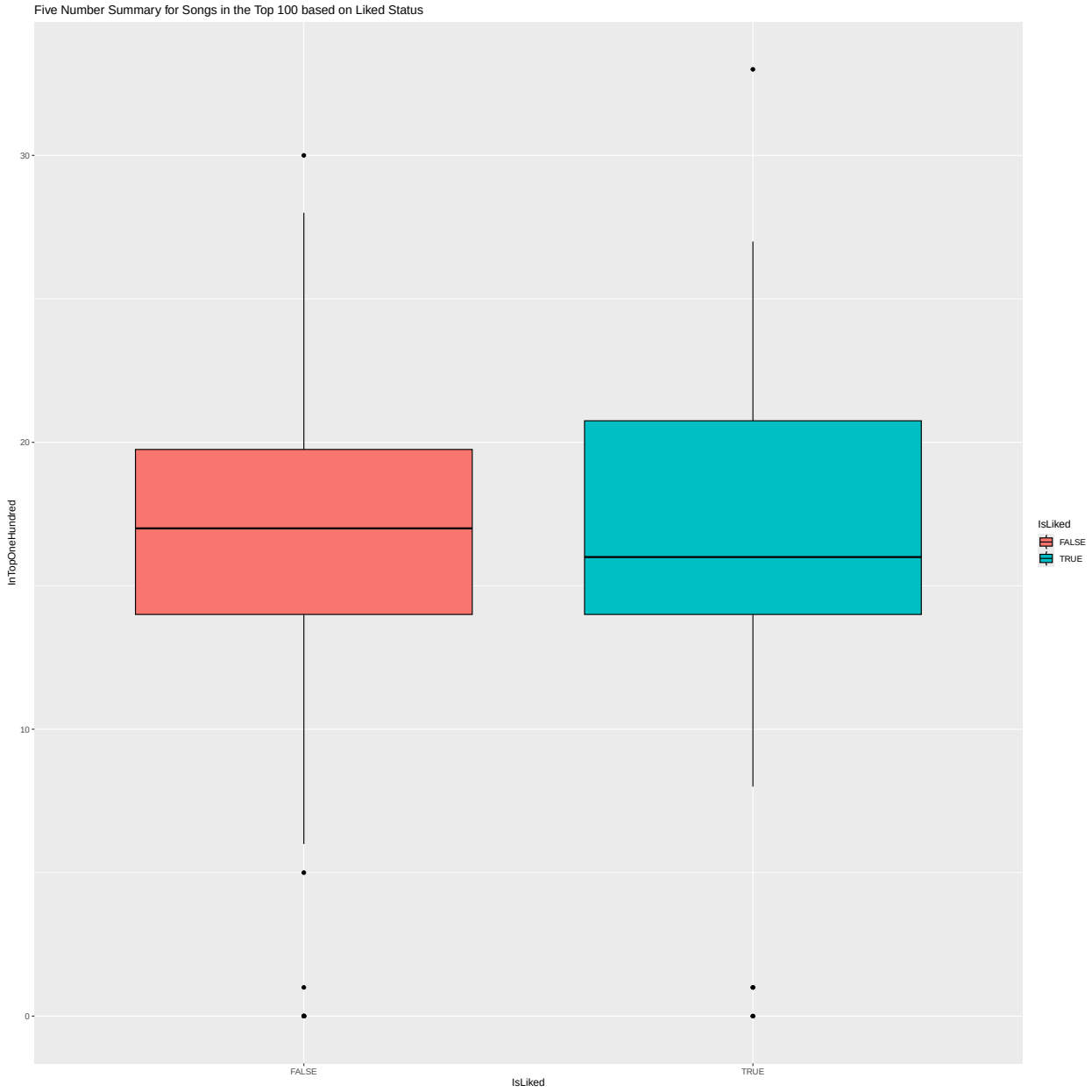

```
ggplot(SongCue, aes(x=SongArtist,y=InTopFifty))+
  geom_col()+
  theme(axis.text.x=element_text(hjust=1,angle=30))+
    labs(title = "Which Bands are Shuffled to the Top Fifty",
       x = "Band Name", y = "Times Shuffled Top Fifty")
```



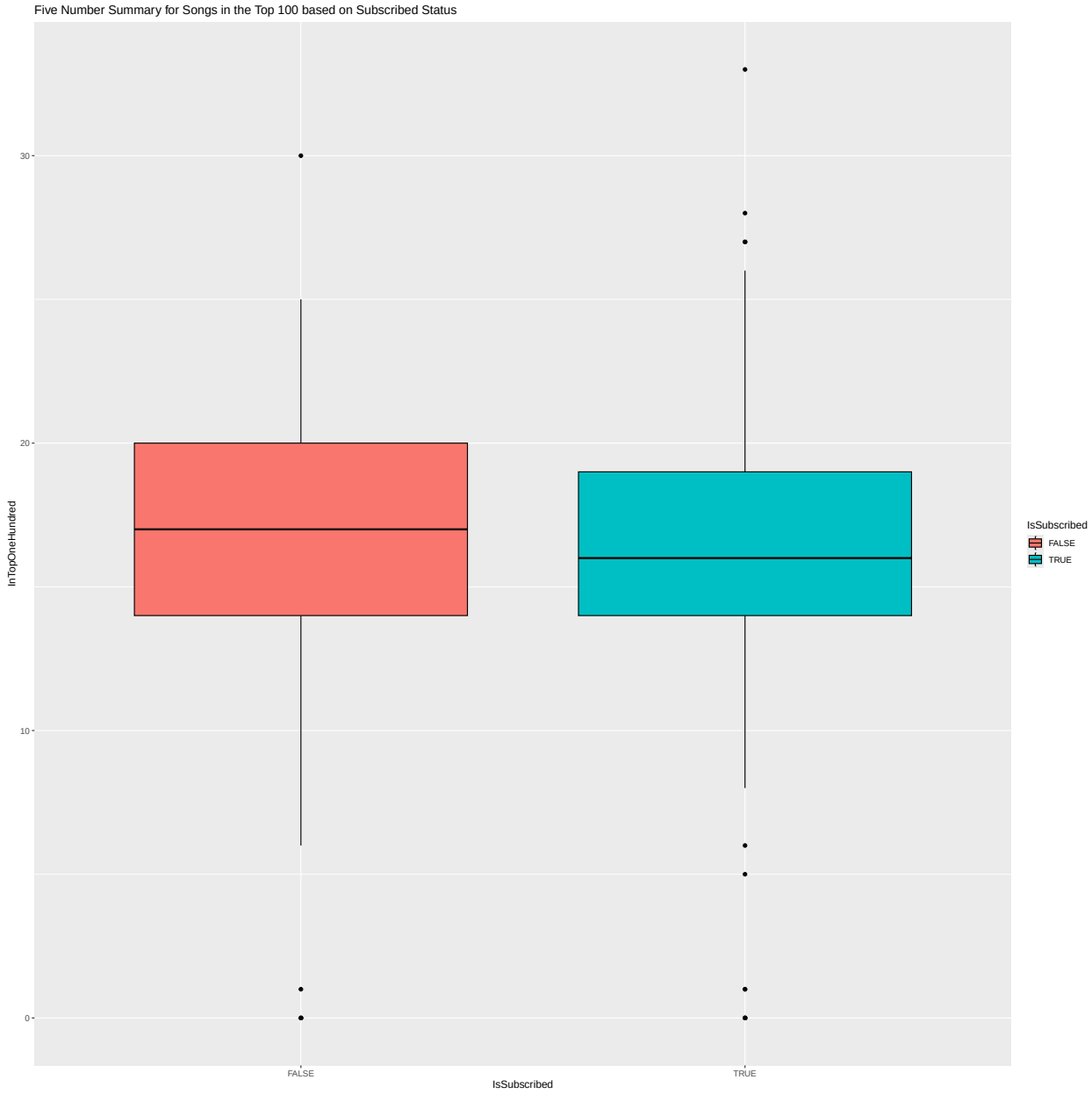

```
ggplot(SongCue, aes(x=SongArtist,y=InTopOneHundred))+
  geom_col(aes(fill=SongArtist,color=SongArtist))+
  theme(axis.text.x=element_text(hjust=1,angle=30))+
    labs(title = "Which Bands are Shuffled to Top One Hundred",
       x = "Band Name", y = "Times Shuffled Top One Hundred")
```



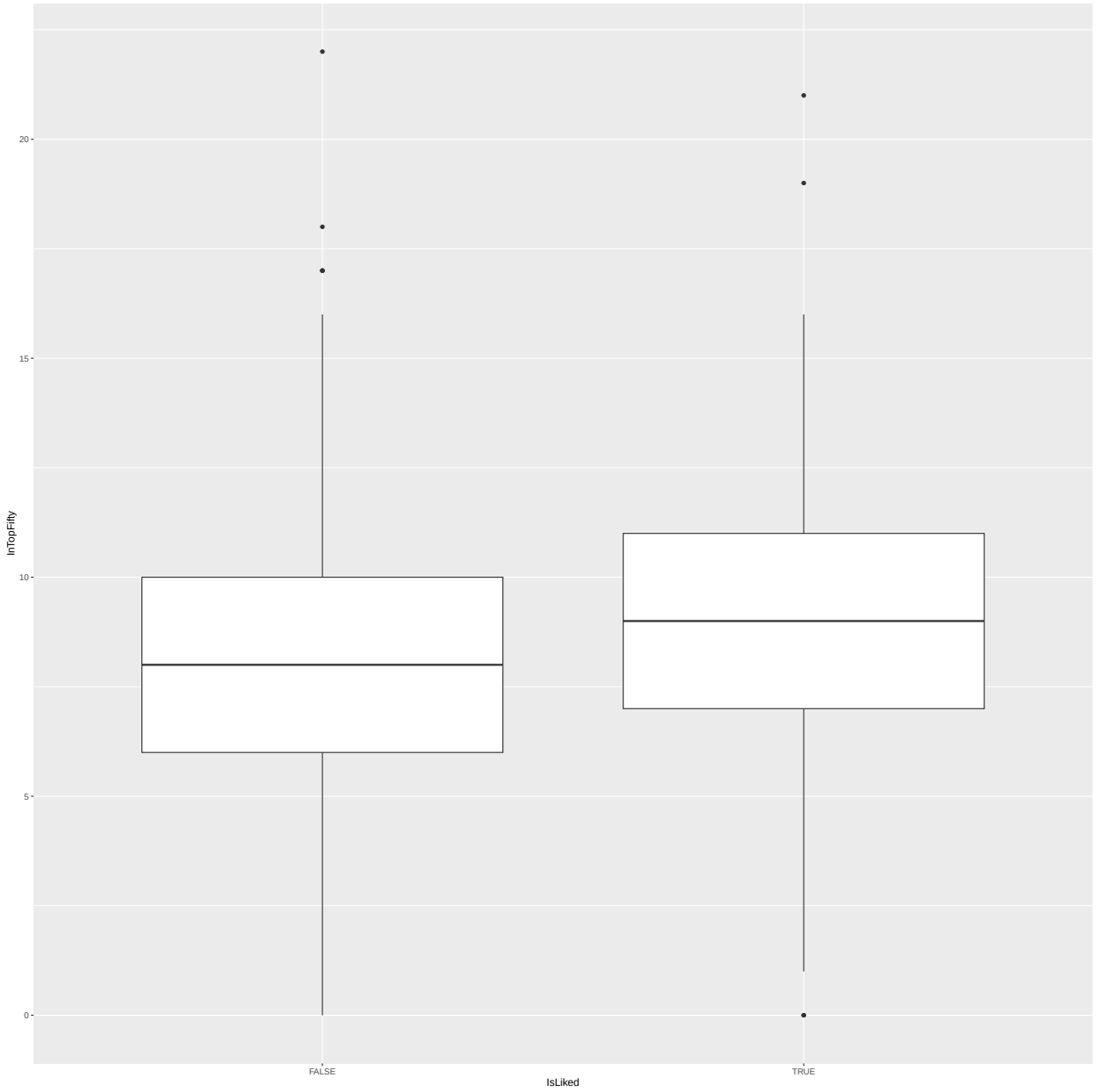

```
ggplot(SongCue, aes(x=IsLiked,y=InTopOneHundred))+
  geom_boxplot(aes(fill=IsLiked),color="BLACK")+
  labs(title = "Five Number Summary for Songs in the Top 100 based on Liked Status")
```



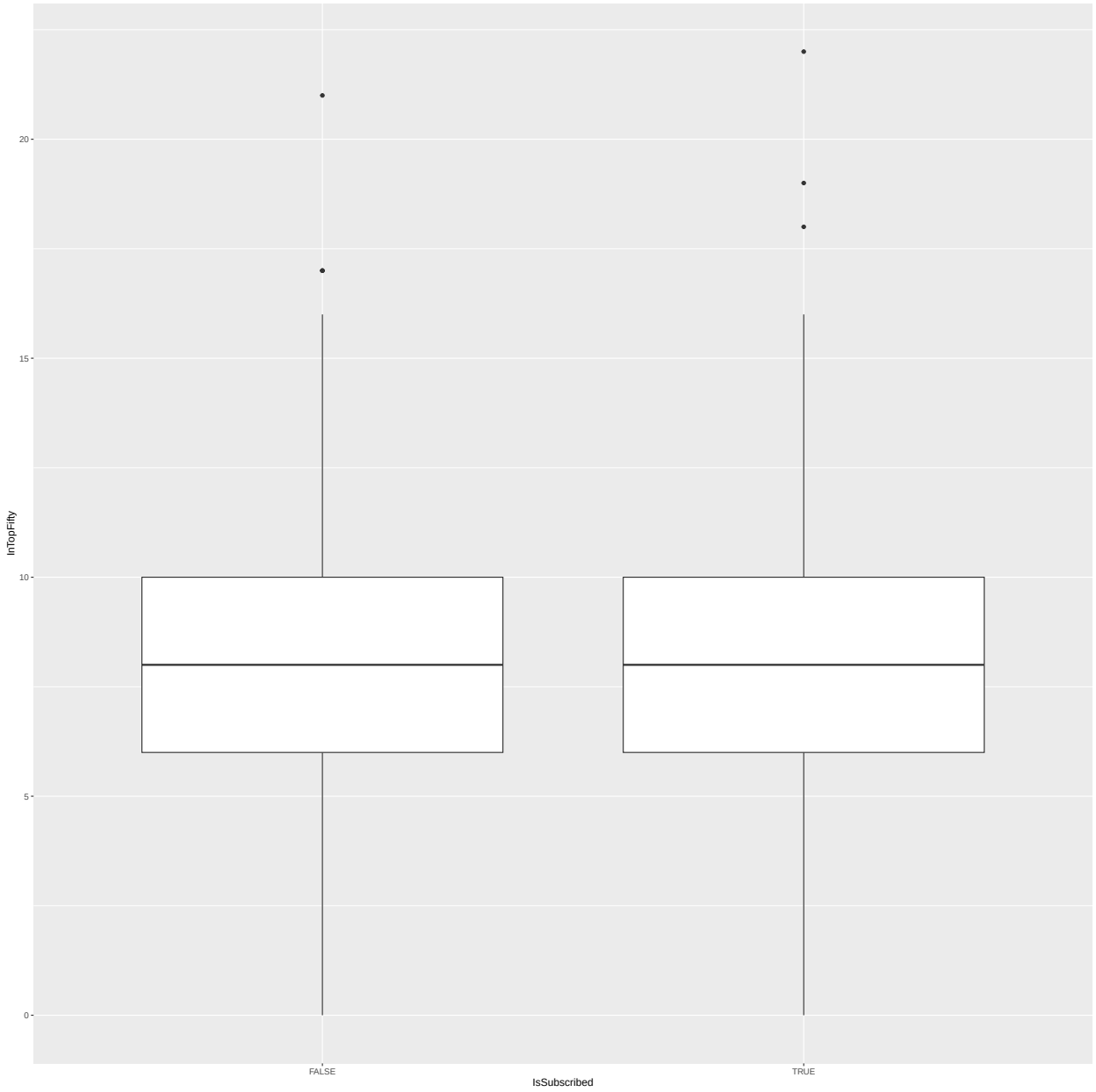

```
ggplot(SongCue, aes(x=IsSubscribed,y=InTopOneHundred))+
  geom_boxplot(aes(fill=IsSubscribed),color="BLACK")+
  labs(title = "Five Number Summary for Songs in the Top 100 based on Subscribed Status")
```



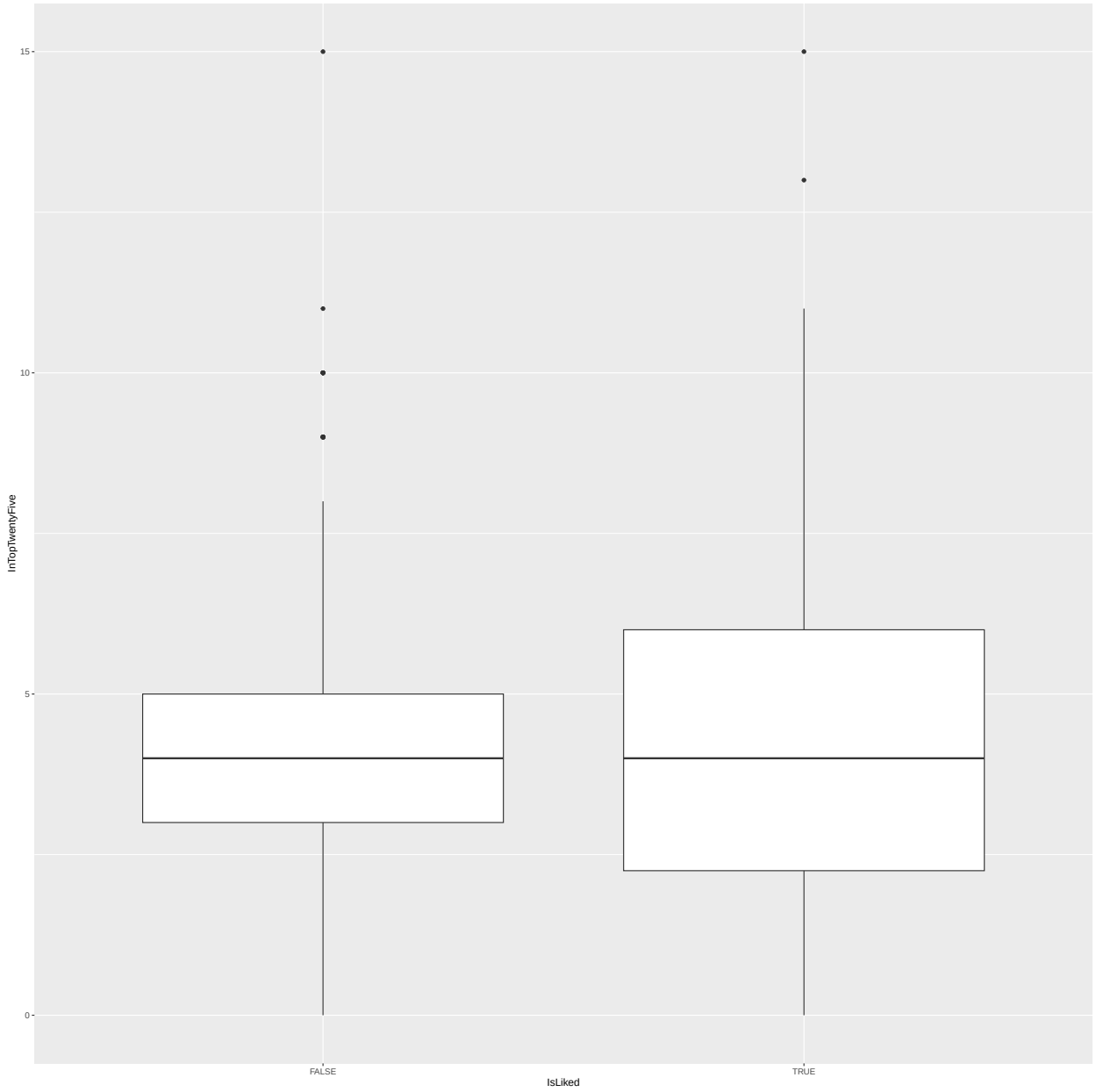

```
ggplot(SongCue, aes(x=IsLiked,y=InTopFifty))+
  geom_boxplot()
```



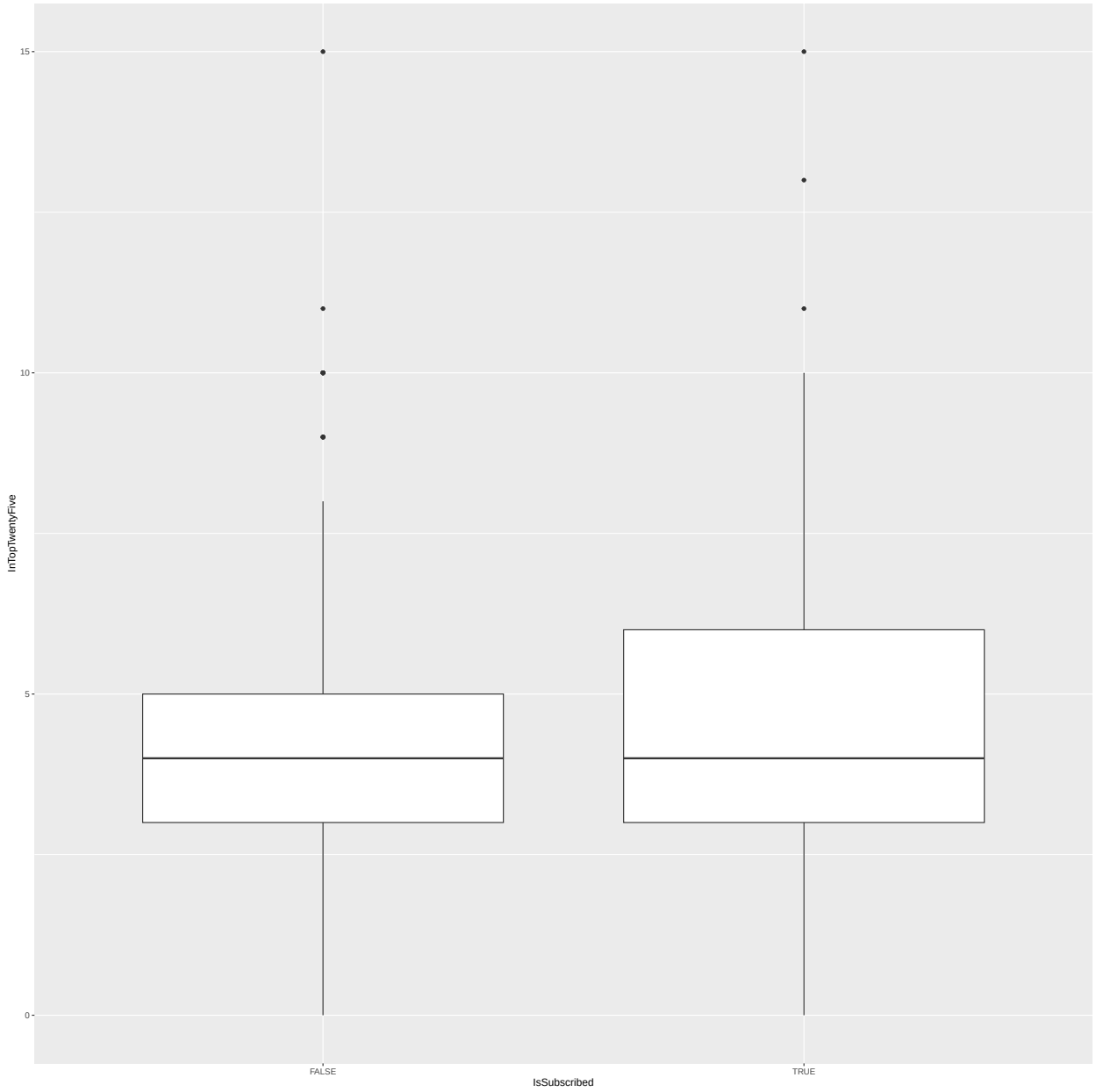

```
ggplot(SongCue, aes(x=IsSubscribed,y=InTopFifty))+
  geom_boxplot()
```



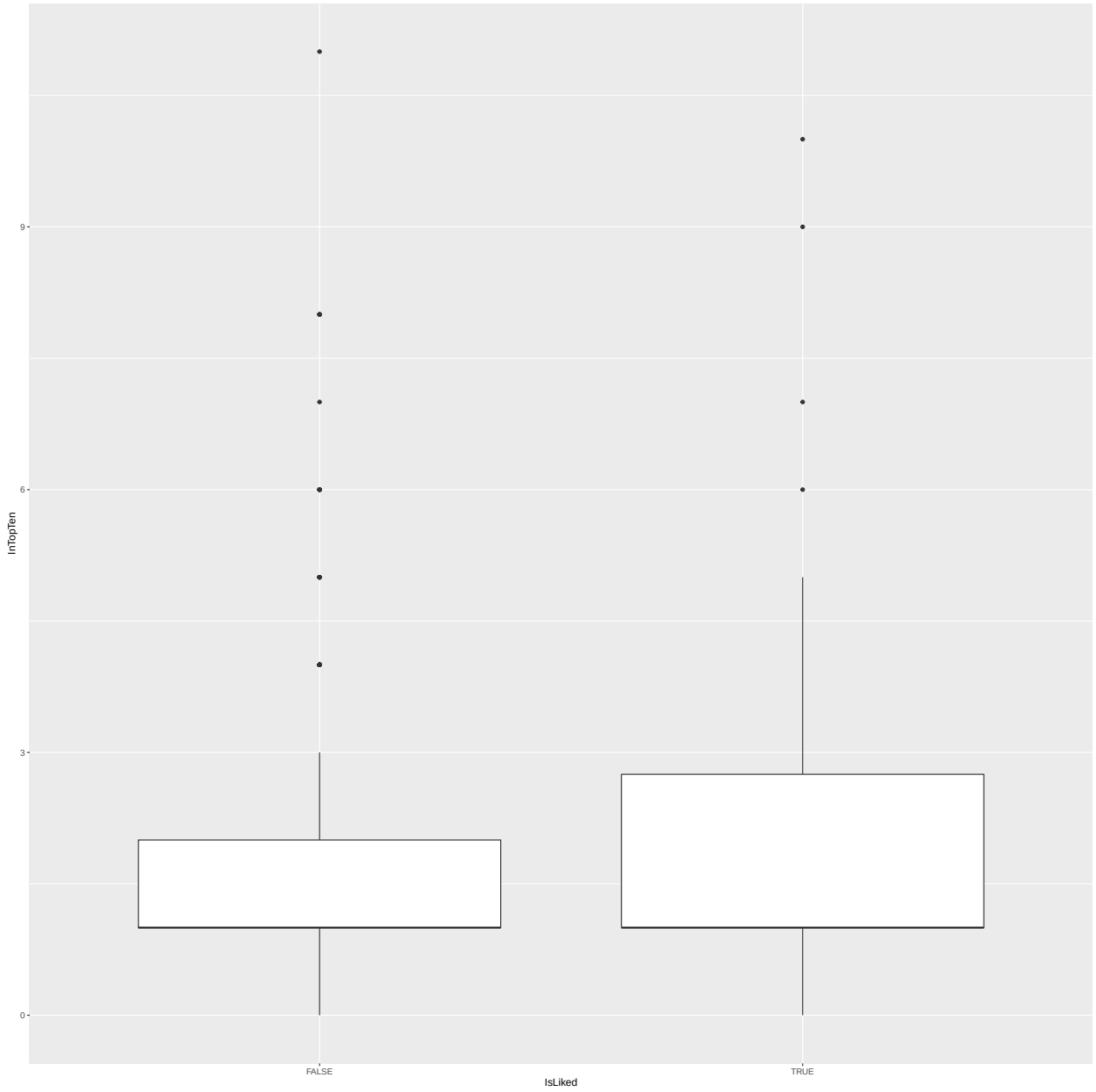

```
ggplot(SongCue, aes(x=IsLiked,y=InTopTwentyFive))+
  geom_boxplot()
```



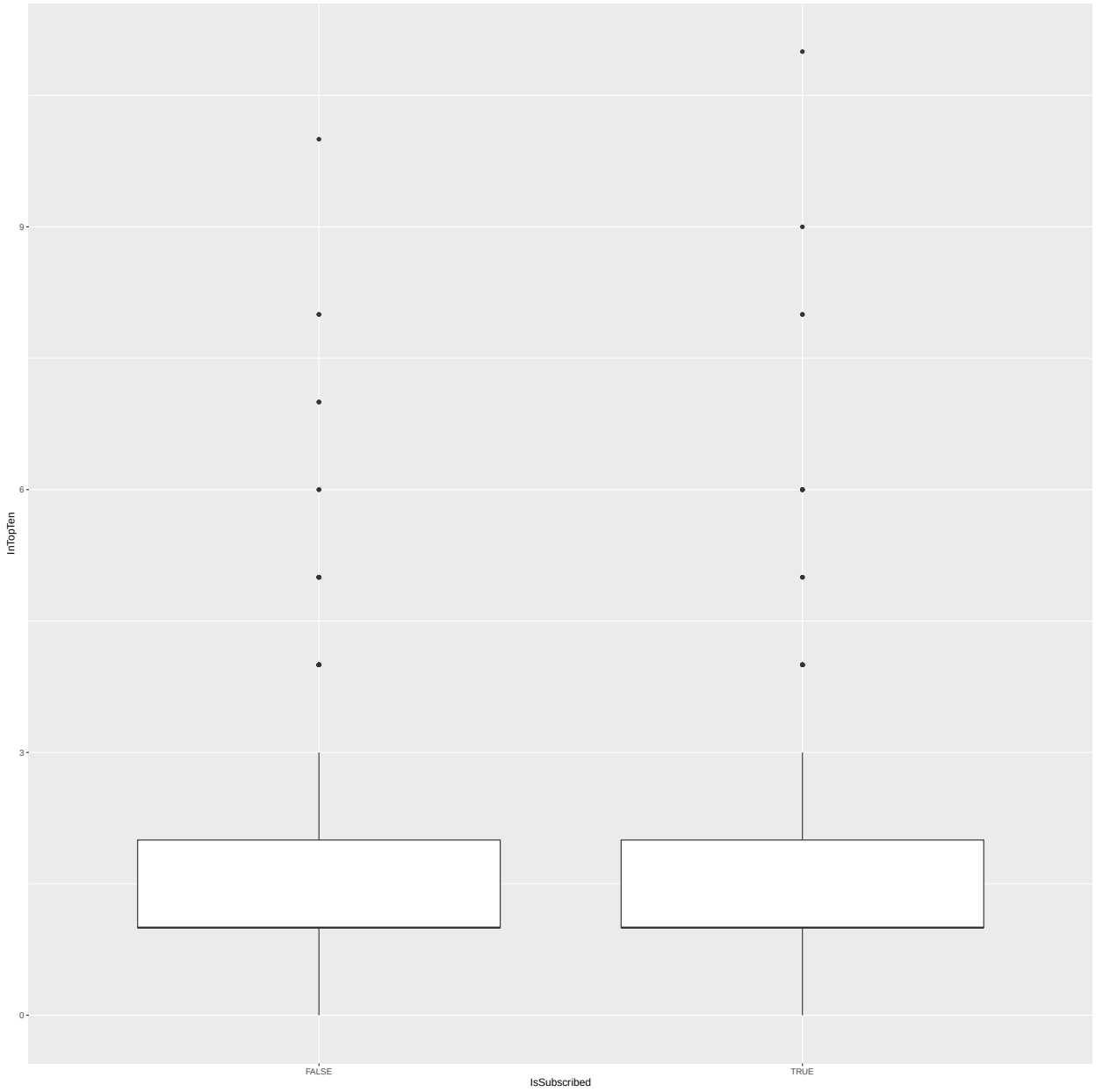

```
ggplot(SongCue, aes(x=IsSubscribed,y=InTopTwentyFive))+
  geom_boxplot()
```



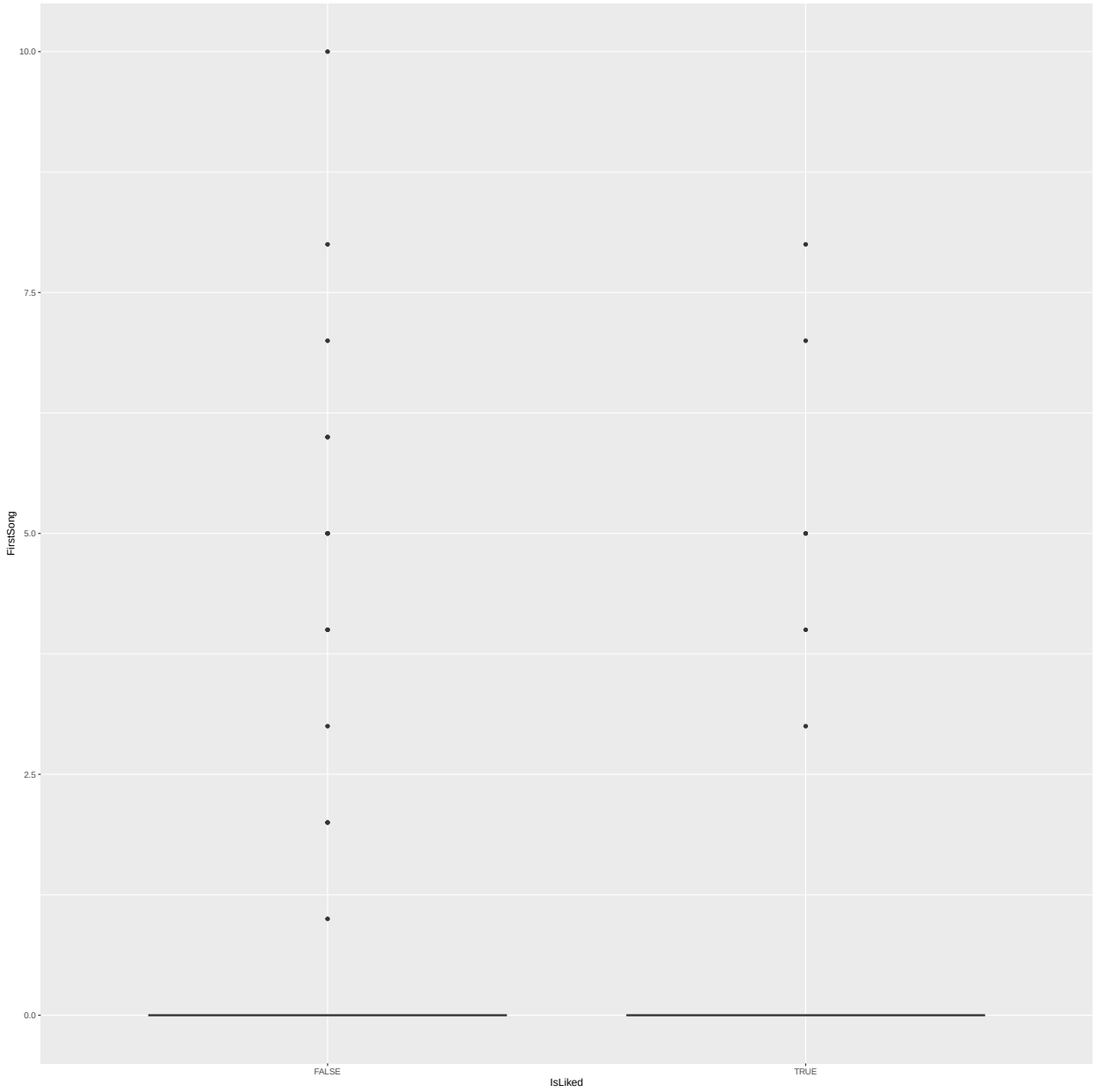

```
ggplot(SongCue, aes(x=IsLiked,y=InTopTen))+
  geom_boxplot()
```



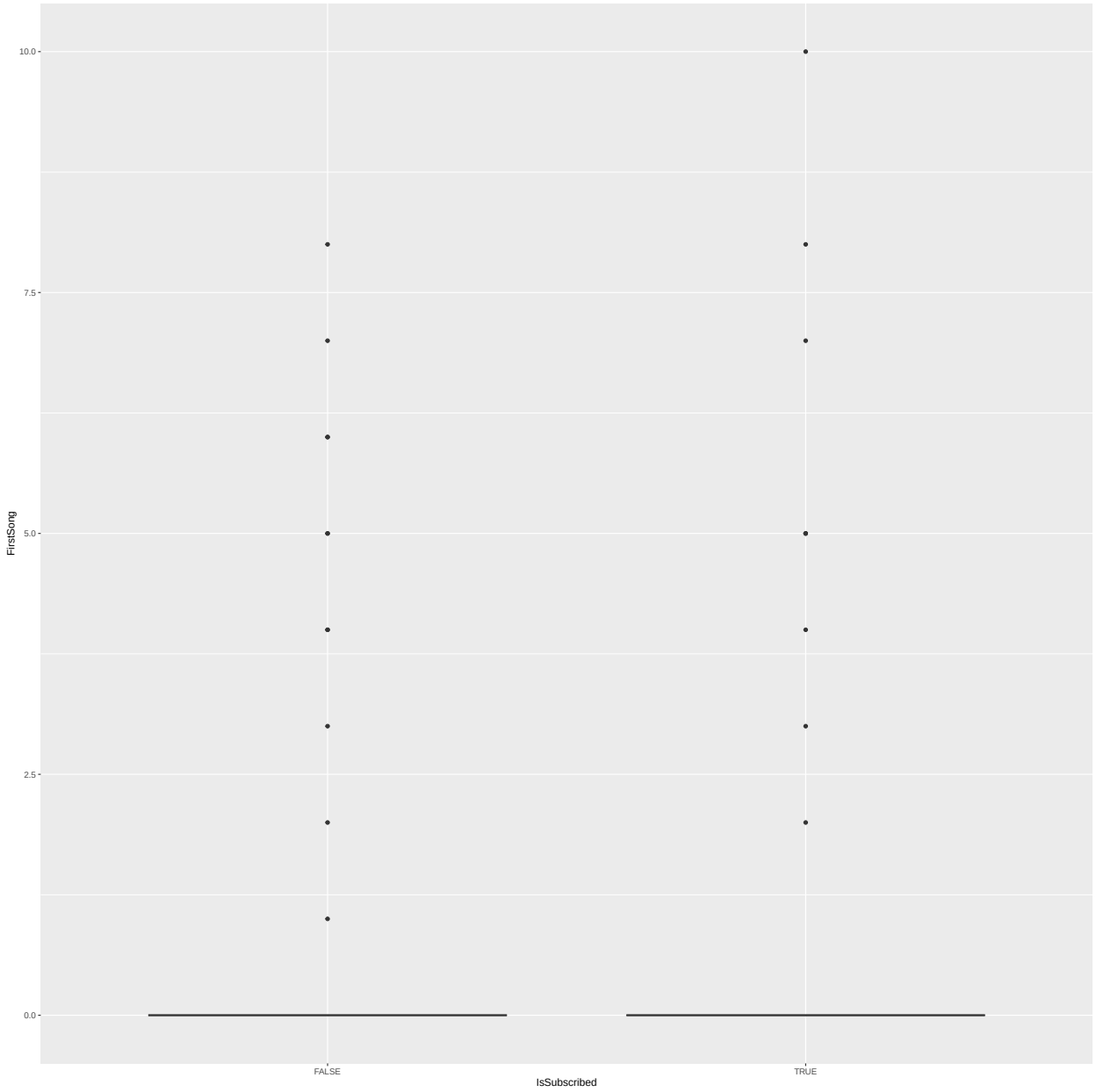

```
ggplot(SongCue, aes(x=IsSubscribed,y=InTopTen))+
  geom_boxplot()
```



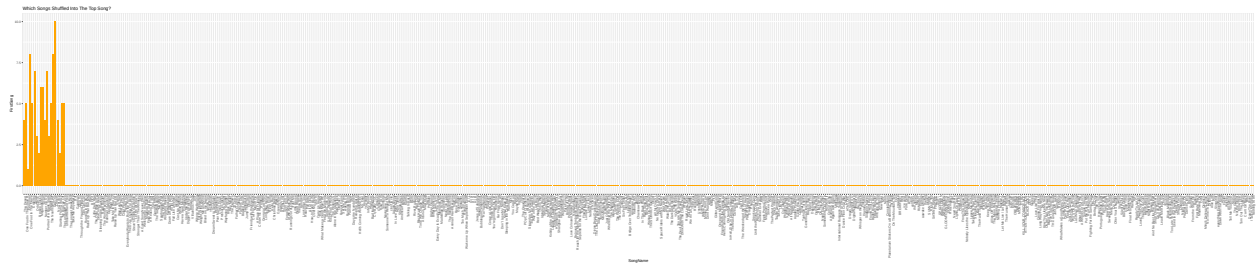

```
ggplot(SongCue, aes(x=IsLiked,y=FirstSong))+
  geom_boxplot()
```



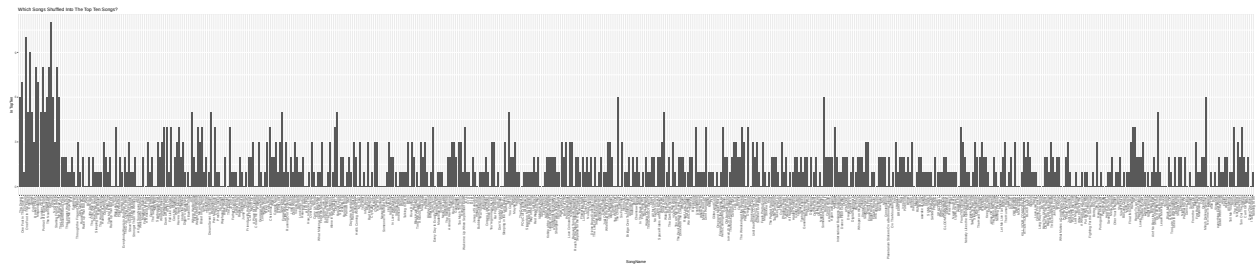

```
ggplot(SongCue, aes(x=IsSubscribed,y=FirstSong))+
  geom_boxplot()
```




```
ggplot(SongCue, aes(x=SongName,y=FirstSong))+
  geom_col(fill="Orange")+
  theme(axis.text.x=element_text(hjust=1,angle=90))+
  scale_x_discrete(limits = SongCue$SongName)+
  labs(title="Which Songs Shuffled Into The Top Song?")
```



```
ggplot(SongCue, aes(x=SongName,y=InTopTen))+
  geom_col()+
  theme(axis.text.x=element_text(hjust=1,angle=90))+
  scale_x_discrete(limits = SongCue$SongName)+
  labs(title="Which Songs Shuffled Into The Top Ten Songs?")
```



```
tableBase<-table(SongCue$SongArtist)
freqBase<-as.vector(tableBase/sum(tableBase))
freqBase<-freqBase*length(SongVectorList)
sumFirstSong<-sum(SongCue$FirstSong)
tableFirstSong<-c(numeric(6))
for(i in 1:length(SongCue$FirstSong)){
  if(SongCue$SongArtist[i]=="Others"){
    tableFirstSong[1]<-tableFirstSong[1]+SongCue$FirstSong[i]
  } else if(SongCue$SongArtist[i]=="Cage The Elephant"){
    tableFirstSong[2]<-tableFirstSong[2]+SongCue$FirstSong[i]
  } else if(SongCue$SongArtist[i]=="PoorMansPoison"){
    tableFirstSong[3]<-tableFirstSong[3]+SongCue$FirstSong[i]
  } else if(SongCue$SongArtist[i]=="The Crane Wives"){
    tableFirstSong[4]<-tableFirstSong[4]+SongCue$FirstSong[i]
  } else if(SongCue$SongArtist[i]=="The Heavy"){
    tableFirstSong[5]<-tableFirstSong[5]+SongCue$FirstSong[i]
  } else {
    tableFirstSong[6]<-tableFirstSong[6]+SongCue$FirstSong[i]
  }
}
freqFirstSong<-tableFirstSong/sum(tableFirstSong)
freqFirstSong<-freqFirstSong*length(SongVectorList)
tableTopTen<-c(numeric(6))
for(i in 1:length(SongCue$InTopTen)){
  if(SongCue$SongArtist[i]=="Others"){
    tableTopTen[1]<-tableTopTen[1]+SongCue$InTopTen[i]
  } else if(SongCue$SongArtist[i]=="Cage The Elephant"){
    tableTopTen[2]<-tableTopTen[2]+SongCue$InTopTen[i]
  } else if(SongCue$SongArtist[i]=="PoorMansPoison"){
```

```
    tableTopTen[3]<-tableTopTen[3]+SongCue$InTopTen[i]
  } else if(SongCue$SongArtist[i]=="The Crane Wives"){
    tableTopTen[4]<-tableTopTen[4]+SongCue$InTopTen[i]
  } else if(SongCue$SongArtist[i]=="The Heavy"){
    tableTopTen[5]<-tableTopTen[5]+SongCue$InTopTen[i]
  } else {
    tableTopTen[6]<-tableTopTen[6]+SongCue$InTopTen[i]
  }
}
freqTopTen<-tableTopTen/sum(tableTopTen)
freqTopTen<-freqTopTen*length(SongVectorList)
tableInTopTwentyFive<-c(numeric(6))
for(i in 1:length(SongCue$SongArtist)){
  if(SongCue$SongArtist[i]=="Others"){
    tableInTopTwentyFive[1]<-tableInTopTwentyFive[1]+SongCue$InTopTwentyFive[i]
  } else if(SongCue$SongArtist[i]=="Cage The Elephant"){
    tableInTopTwentyFive[2]<-tableInTopTwentyFive[2]+SongCue$InTopTwentyFive[i]
  } else if(SongCue$SongArtist[i]=="PoorMansPoison"){
    tableInTopTwentyFive[3]<-tableInTopTwentyFive[3]+SongCue$InTopTwentyFive[i]
  } else if(SongCue$SongArtist[i]=="The Crane Wives"){
    tableInTopTwentyFive[4]<-tableInTopTwentyFive[4]+SongCue$InTopTwentyFive[i]
  } else if(SongCue$SongArtist[i]=="The Heavy"){
    tableInTopTwentyFive[5]<-tableInTopTwentyFive[5]+SongCue$InTopTwentyFive[i]
  } else {
    tableInTopTwentyFive[6]<-tableInTopTwentyFive[6]+SongCue$InTopTwentyFive[i]
  }
}
freqInTopTwentyFive<-tableInTopTwentyFive/sum(tableInTopTwentyFive)
freqInTopTwentyFive<-freqInTopTwentyFive*length(SongVectorList)
tableInTopFifty<-c(numeric(6))
for(i in 1:length(SongCue$SongArtist)){
  if(SongCue$SongArtist[i]=="Others"){
    tableInTopFifty[1]<-tableInTopFifty[1]+SongCue$InTopFifty[i]
  } else if(SongCue$SongArtist[i]=="Cage The Elephant"){
    tableInTopFifty[2]<-tableInTopFifty[2]+SongCue$InTopFifty[i]
  } else if(SongCue$SongArtist[i]=="PoorMansPoison"){
    tableInTopFifty[3]<-tableInTopFifty[3]+SongCue$InTopFifty[i]
  } else if(SongCue$SongArtist[i]=="The Crane Wives"){
    tableInTopFifty[4]<-tableInTopFifty[4]+SongCue$InTopFifty[i]
  } else if(SongCue$SongArtist[i]=="The Heavy"){
    tableInTopFifty[5]<-tableInTopFifty[5]+SongCue$InTopFifty[i]
  } else {
    tableInTopFifty[6]<-tableInTopFifty[6]+SongCue$InTopFifty[i]
  }
}
freqInTopFifty<-tableInTopFifty/sum(tableInTopFifty)
freqInTopFifty<-freqInTopFifty*length(SongVectorList)
tableInTopOneHundred<-c(numeric(6))
for(i in 1:length(SongCue$SongArtist)){
  if(SongCue$SongArtist[i]=="Others"){
    tableInTopOneHundred[1]<-tableInTopOneHundred[1]+SongCue$InTopOneHundred[i]
  } else if(SongCue$SongArtist[i]=="Cage The Elephant"){
    tableInTopOneHundred[2]<-tableInTopOneHundred[2]+SongCue$InTopOneHundred[i]
```

```
  } else if(SongCue$SongArtist[i]=="PoorMansPoison"){
    tableInTopOneHundred[3]<-tableInTopOneHundred[3]+SongCue$InTopOneHundred[i]
  } else if(SongCue$SongArtist[i]=="The Crane Wives"){
    tableInTopOneHundred[4]<-tableInTopOneHundred[4]+SongCue$InTopOneHundred[i]
  } else if(SongCue$SongArtist[i]=="The Heavy"){
    tableInTopOneHundred[5]<-tableInTopOneHundred[5]+SongCue$InTopOneHundred[i]
  } else {
    tableInTopOneHundred[6]<-tableInTopOneHundred[6]+SongCue$InTopOneHundred[i]
  }
}
freqInTopOneHundred<-tableInTopOneHundred/sum(tableInTopOneHundred)
freqInTopOneHundred<-freqInTopOneHundred*length(SongVectorList)
```

```
chiFirstSong<-0
chiTopTen<-0
chiTopTwentyFive<-0
chiTopFifty<-0
chiTopHundred<-0
freqBase<-as.vector(freqBase)
for(ind in 1:length(freqFirstSong)){
  chiFirstSong<-chiFirstSong+(((freqFirstSong[ind]-freqBase[ind])^2)/freqBase[ind])
}
for(ind in 1:length(freqTopTen)){
  chiTopTen<-chiTopTen+(((freqTopTen[ind]-freqBase[ind])^2)/freqBase[ind])
}
for(ind in 1:length(freqInTopTwentyFive)){
  chiTopTwentyFive<-chiTopTwentyFive+(((freqInTopTwentyFive[ind]-freqBase[ind])^2)/freqBase[ind])
}
for(ind in 1:length(freqInTopFifty)){
  chiTopFifty<-chiTopFifty+(((freqInTopFifty[ind]-freqBase[ind])^2)/freqBase[ind])
}
for(ind in 1:length(freqInTopOneHundred)){
  chiTopHundred<-chiTopHundred+(((freqInTopOneHundred[ind]-freqBase[ind])^2)/freqBase[ind])
}
pFirst<-pchisq(chiFirstSong,5,lower.tail = FALSE)
pFirst
```

```
## [1] 2.432355e-19
```

```
pFirst>.05
```

```
## [1] FALSE
```

```
pTen<-pchisq(chiTopTen,5,lower.tail = FALSE)
pTen
```

```
## [1] 0.8174775
```

```
pTen>.05
```

```
## [1] TRUE
```

```r
pTFive<-pchisq(chiTopTwentyFive,5,lower.tail = FALSE)
pTFive
```

```
## [1] 0.9934875
```

```r
pTFive>.05
```

```
## [1] TRUE
```

```r
pFifty<-pchisq(chiTopFifty,5,lower.tail=FALSE)
pFifty
```

```
## [1] 0.9983078
```

```r
pFifty>.05
```

```
## [1] TRUE
```

```r
pHundred<-pchisq(chiTopHundred,5,lower.tail=FALSE)
pHundred
```

```
## [1] 0.9997294
```

```r
pHundred>.05
```

```
## [1] TRUE
```